复旦数学资料汇编
广义相对论

来源：香蕉空间 Solution 资料库

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编者说明

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第一章　广义相对论

1.1　试卷: 广义相对论

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2024–2025 Semester 1 数学广义相对论 (谢纳庆)

1. (30 Points) (1) We assume that ξ^η is a Killing field, that is,

$$\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha = 0. \tag{1.1}$$

We also assume that c is a timelike geodesic with unit tangent vector p^α. Show that along c, $p^\alpha \xi_\alpha$ is constant.

(2) Give 10 examples of Killing fields of Minkowski space $\mathbb{R}^{3,1}$.

2. (30 Points) For the Schwarzschild solution

$$g = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(-\frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \tag{1.2}$$

the Komar mass coincides with the ADM mass.

3. (40 Points) (1) If all the $g_{\alpha\beta}$'s are independent of the fixed x^μ, show that the equation of the energy momentum tensor $T^\nu_{\ \mu;\nu} = 0$ is equivalent to

$$\frac{1}{\sqrt{-g}}(\sqrt{-g}T^\nu_{\ \mu})_{,\nu} = 0. \tag{1.3}$$

(2) Assume further that for each fixed x_0, the set of (x_1, x_2, x_3) such that $T^\nu_{\ \mu} \neq 0$ for some μ, ν is bounded. Show with equation (3) that

$$\int_{x^0=\text{const}} T^\nu_{\ \mu}\sqrt{-g}n_\nu d^3x \tag{1.4}$$

is independent of x^0, where n_ν is the unit cotangent vector of the constant time surface.

(3) Consider the Minkowski spacetime with polar coordinates (t, r, θ, ϕ), show with equation (4) that

$$J = \int_{t=\text{const}} T^0_{\ \phi} r^2 \sin\theta dr d\theta d\phi \tag{1.5}$$

is independent of t.

(4) Change equation (5) into the rectangular coordinates (t, x, y, z) to conclude that

$$J = \int (xT^{y0} - yT^{x0}) dx dy dz \tag{1.6}$$

is independent of t.

2025-2026 Semester 2 广义相对论 (PHYS60019)

1.Consider a 3D spacetime with the following metric

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2\left(\frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\phi^2\right),$$

where $a = a(t)$ is a function of t and k is a constant.

(a) (5 points) Calculate all non-vanishing Christoffel symbols:

$$\Gamma^{\mu}_{\nu\rho} = \frac{1}{2}g^{\mu\sigma}(\partial_\nu g_{\sigma\rho} + \partial_\rho g_{\nu\sigma} - \partial_\sigma g_{\nu\rho}).$$

(b) (5 points) Write the Lagrangian of a free point-like massive particle in this metric.

(c) (5 points) Write the constants of motion of a free point-like massive particle in this metric.

(d) (5 points) Calculate the relation between the proper time of a static observer (r and ϕ constant) and the time coordinate t.

(e) (5 points) Calculate the relation between the proper time of an observer moving along a circle of radius R with angular frequency Ω ($\Omega = \mathrm{d}\phi/\mathrm{d}t$) and the time coordinate t.

(f) (5 points) Write the vierbeins $E^{\mu}_{(\alpha)}$ to move to a locally Minkowski reference frame (i.e., $dx^\mu \to d\hat{x}^{(\alpha)} = E^{(\alpha)}_{\mu}dx^\mu$ such that $g_{\mu\nu} \to \eta_{(\alpha)(\beta)}$).

(g) (10 points) Let us now assume the specific case $a = 3$ (independent of t) and $k = 0$. Write all non-vanishing coefficients of the Ricci tensor:

$$R_{\mu\nu} = \partial_\lambda\Gamma^{\lambda}_{\mu\nu} - \partial_\nu\Gamma^{\lambda}_{\mu\lambda} + \Gamma^{\kappa}_{\mu\nu}\Gamma^{\lambda}_{\kappa\lambda} - \Gamma^{\kappa}_{\mu\lambda}\Gamma^{\lambda}_{\nu\kappa}.$$

(h) (5 points) Still assuming $a = 3$ and $k = 0$, calculate the Einstein tensor: $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$.

2.(15 points)The field equation for a scalar field ϕ in Cartesian coordinates in flat spacetime is (in units $\hbar = c = 1$)

$$(\Box - m^2)\phi = 0,$$

where $\Box = \partial_\mu\partial^\mu = -\partial_t^2 + \partial_x^2 + \partial_y^2 + \partial_z^2$ and m is the mass of the scalar field.

Write this equation in spherical coordinates.

3.(20 points)The Global Positioning System (GPS) is the most famous global navigation satellite system. It consists of a constellation of satellites orbiting at an altitude $h \approx 20,000$ km from the ground and with an orbital speed $v \approx 14,000$ km/hr. GPS satellites carry stable atomic clocks and continually broadcast a signal that includes their current time and position. From the signal of at least four satellites, a GPS receiver on Earth ($h \approx 0$ km) can determine its position.

Calculate the difference between the time of the clock of the GPS receiver and the time of the clocks of the GPS satellites after 24 hr.

4.(20 points)Consider a Cartesian coordinate system of an inertial reference frame in which we measure a constant uniform electric field $\mathbf{E} = (E, 0, 0)$.

Calculate the electric field measured in the inertial reference frame moving with constant velocity $\mathbf{v} = (v, 0, 0)$ with respect to the former.

2025–2026 Semester 2 广义相对论期末 (PHYS60019)

1. (16 Points) The Penrose diagram appended below describes an asymptotically flat spacetime in which a star S collapses and forms a black hole, and then the black hole completely evaporates by Hawking radiation.

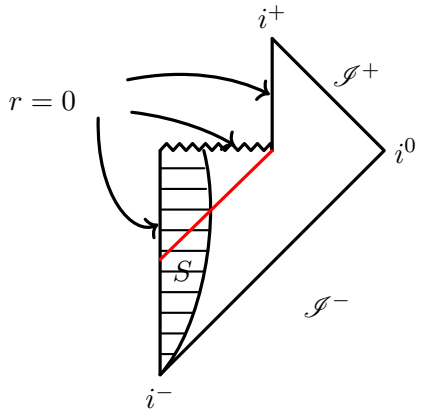


Figure 1: Penrose diagram for Question 1

(1) Indicate clearly the 5 asymptotic regions: future time-like infinity, past time-like infinity, spatial infinity, future null infinity, past null infinity. [8 Points]

(2) Indicate clearly the black hole horizon and the singularity of the spacetime. [8 Points]

2. (24 Points) Let us now consider the Penrose diagram appended below.

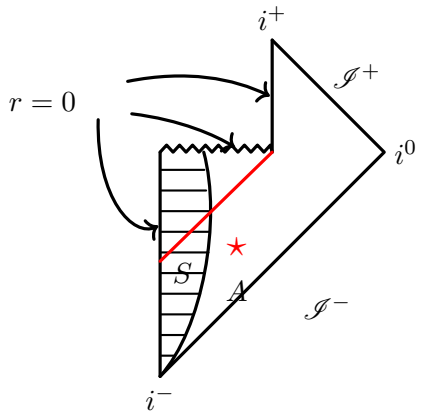


Figure 2: Penrose diagram for Question 2

(1) We consider the coordinate system of an observer at infinity (t, r, θ, ϕ). At time $t = t_0$, a spacecraft is at point A. Draw the trajectory of the spacecraft from $t = t_0$ to $t = \infty$ assuming that the spacecraft travels at the constant velocity $v = 0.999999999c$ ($c =$ speed of light) with $dr/dt > 0$. Is the trajectory of the spacecraft a time-like geodesic? [8 Points]

(2) Let us now assume that at time $t = t_0$ the spacecraft is still at point A but it is orbiting around the star S. The orbit is circular $(dr/dt = 0)$ and remains circular. Draw the trajectory of the spacecraft from $t = t_0$ to $t = \infty$. [8 Points]

(3) Suppose that the spacecraft at point A emits an electromagnetic signal. Draw the trajectory of such an electromagnetic signal. [8 Points]

3. (60 Points) Let us consider the following line element

$$ds^2 = -dt^2 + a^2\left(\frac{dr^2}{1-r^2} + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2\right), \tag{1.7}$$

where $a = a(t) = \sqrt{t - t^2}$ is a function of t.

(1) Is this a universe filled with matter, radiation, or vacuum energy? Please explain how you can infer the matter content or why you cannot answer this question. [10 Points]

(2) Write the energy E and the axial component of the angular momentum L_z of a test-particle in this spacetime. Are E and L_z constants of motion? [5 Points]

(3) Assuming that the spacetime has a trivial topology, is the 3-volume finite or infinite? [5 Points]

(4) At the time $t = 0.1$, an atom emits a photon of energy $E = 5\,\text{eV}$. What is the photon energy at the time $t = 0.3$? Can a photon be blueshifted in this spacetime and why? [8 Points]

(5) Does the spacetime have an ergoregion? In the case of positive answer, where? [8 Points]

(6) Does the spacetime have curvature singularities? In the case of positive answer, where? [8 Points]

(7) Write the vierbeins $E^{\mu}{}_{(\alpha)}$. [8 Points]

(8) Is the spacetime asymptotically flat? In the case of positive answer, draw its Penrose diagram. [8 Points]

1.2　课程概要及预备知识

原网页

comment| 这是 2025 春季学期谢纳庆老师数学广义相对论课程概要及预备知识。

本课程教材为 Jose Natario 的 *An Introduction to Mathematical Relativity*，基本按照教材讲述。这里快速介绍一下狭义相对论和微分几何的预备知识，并不完整。若有需求可以阅读参考书梁灿彬、周彬的《微分几何入门与广义相对论》上册的对应章节。

1.3　*Preliminaries*

The main topics of this course.

(1) Penrose’s singularity theorem.

(2) Cauchy problem of Einstein Equation.

(3) Positive Theorem .

1.3.1　Introduction

最重要的 Equation 被希望是在惯性坐标系下保持不变的。

Example 1.3.1. *Newtonian 2nd law.*

$$F = m \cdot a = m \cdot \frac{d^2x}{dt^2}$$

Galileo transformation.

$$\begin{cases} x' = x - vt \\ t' = t \end{cases}$$

where v is a constant velocity vector.

$$F = m \cdot \frac{d^2 x'}{dt'^2}$$

Source-free Maxwell equation in vacuum.

$$\begin{cases} \vec{\nabla} \cdot \vec{E} = 0 \\ \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \cdot \vec{B} = 0 \\ \vec{\nabla} \times \vec{B} = \frac{\partial \vec{E}}{\partial t} \end{cases}$$

$$\frac{\partial^2 \vec{E}}{\partial t^2} = \frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B}) = \cdots$$

So we can get

$$\begin{cases} \frac{\partial^2 \vec{E}}{\partial t^2} - \Delta \vec{E} = 0, \\ \frac{\partial^2 \vec{B}}{\partial t^2} - \Delta \vec{B} = 0. \end{cases}$$

注 1.3.1. 这门课都采用 *Gauss* 单位制，光速 $c = 1$.

The Galieo transformation does not preserve the form of the Maxwell equation.

$\rightarrow$ It turns out that the Lorentz group preserves the form of Maxwell equation.

$x, y \in R^3$, assume that $A \in SO(3)$, then

$$\langle Ax, Ay \rangle = (Ax)^T (Ay) = x^T A^T A y = x^T y.$$

Let $R^{3,1}$ be the so called Minkowski spacetime, i.e., $R^{3,1}$ is diffeomorphic to R^4 and is endowed with a symmetric, nondegenerate, bilinear 2-form

$$g = -dt^2 + \sum_{i=1}^{3} (dx^i)^2 = (dt, dx^1, dx^2, dx^3) \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \begin{pmatrix} dt \\ dx^1 \\ dx^2 \\ dx^3 \end{pmatrix}$$

Denoted by $\eta = \text{diag}(-1, 1, 1, 1)$.

If we take a Lorentz transformation

$$\begin{cases} x' = \gamma(x - vt) \\ t' = \gamma(t - vx) \\ y' = y \\ z' = z \end{cases}$$

where v is constant and $\gamma = \frac{1}{\sqrt{1-v^2}}$.

Fact: The Maxwell equation is invariant under the change of coordinates above.

$$\begin{pmatrix} t' \\ \\ x' \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1-v^2}} & \frac{-v}{\sqrt{1-v^2}} \\ \\ \frac{-v}{\sqrt{1-v^2}} & \frac{1}{\sqrt{1-v^2}} \end{pmatrix} \begin{pmatrix} t \\ \\ x \end{pmatrix}$$

The Lorentz group is the symmetry group of the Maxwell equation.

If $v \ll 1$, then $y \approx 1$.

$$\frac{dx'}{dt'} = \frac{y(dx - vdt)}{y(dt - vdx)} = \frac{\frac{dx}{dt} - v}{1 - v\frac{dx}{dt}}$$

In particular, $\frac{dx}{dt} = 1 \implies \frac{dx'}{dt'} = 1$. (光速不变)

Denote by $\eta = (dt, dx^1, dx^2, dx^3) \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \begin{pmatrix} dt \\ dx^1 \\ dx^2 \\ dx^3 \end{pmatrix}$

the Lorentz transformation preserves η.

注 **1.3.2.** *If we denote by η the most positive sign form, then it is called the Landau-Lifshitz convention, Geometers prefer this convention. If we make use of the most minus sign convention $(1, -1, -1, -1)$, particle physicists prefer this convention.*

We call η the Minkowski metric

We introduce a skew-symmetric matrix,

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ \\ E_1 & 0 & B_3 & -B_2 \\ \\ E_2 & -B_3 & 0 & B_1 \\ \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

Then the Maxwell equation can be written as

$$\begin{cases} dF = 0 \\ d * F = 0 \end{cases}$$

where d is the extension differential and $*$ is the Hodge star.

Ex. The Lorentz force $\vec{F}$.

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

How to rewrite the Lorentz force in 4-D?

Ex. How about the transformation of $\vec{E}$ and $\vec{B}$ under Lorentz change?

1.3.2 Some Topics about Special Relativity

定义 1.3.1. *A vector v is said to be*

(1) Timelike if $\langle v, v\rangle < 0$.

(2) Spacelike if $\langle v, v\rangle > 0$.

(3) Lightlike or null if $\langle v, v\rangle = 0$.

(4) Causal if it is timelike or null.

(5) Future-pointing if it is causal and $\langle v, \frac{\partial}{\partial t}\rangle < 0$

(6) Past-pointing if $-v$ is future-pointing.

注 1.3.3. *The same classification applies to curves: $c : [a, b] \to \mathbb{R}^{3,1}$. According to its tangential vector.*

Let $c : [a, b] \to \mathbb{R}^{3,1}$ be a timelike curve then $\tau(c) = \int_a^b \sqrt{-\langle \dot{c}(s), \dot{c}(s)\rangle}\, ds$ represents the proper time (固有时) measured by the particle between events $c(a)$ and $c(b)$.

Prop 1.3.1 (Twin paradox)**.** *Of all timelike curves, connecting two events $p, q \in \mathbb{R}^{3,1}$, the curve with maximal length is the line segment (representing initial motion).*

Prop 1.3.2 (Popper Effect)**.** *An observer moving with velocity v away from a source of light of Period T, measure the period to be*

$$T' = T\sqrt{\frac{1+v}{1-v}}$$

证明. These signals are detected by an observer moving with velocity v who measures a time difference T'. Now if the first signal is emitted at t_0, its history is the time $t = t_0 + x$.

Consequently the moving observer defects the signal at the event with coordinates.

$$\begin{cases} t = t_0 + x \\ x = vt \end{cases} \iff \begin{cases} t = \frac{t_0}{1-v} \\ x = \frac{vt_0}{1-v} \end{cases}$$

Similarly, the second light signal is emitted at $t = t_0 + T$, its history is the time $t = t_0 + T + x$.

$$\begin{cases} t = t_0 + T + x \\ x = vt \end{cases} \iff \begin{cases} t = \frac{t_0+T}{1-v} \\ x = \frac{v(t_0+T)}{1-v} \end{cases}$$

$$\Rightarrow T' = \sqrt{(\frac{t_0+T}{1-v} - \frac{t_0}{1-v})^2 - (\frac{v(t_0+T)}{1-v} - \frac{vt_0}{1-v})^2} = T\sqrt{\frac{1+v}{1-v}}. \qquad \square$$

注 1.3.4. $T' > T \Leftrightarrow$ *frequency* $\omega' < \omega$.

这是红移 *(redshift)* 现象，是多普勒效应的体现。

1.4 Review of Differential Geometry

Assume that M is a C^∞ manifold

$$M \cong \mathbb{R}^4.$$

Let g be a Lorentz metric on M.

$$\forall x \in M, \left(g^{(x)}\right)_{\mu\nu} = \begin{pmatrix} g_{00} & g_{01} & g_{02} & g_{03} \\ g_{10} & g_{11} & g_{12} & g_{13} \\ g_{20} & g_{21} & g_{22} & g_{23} \\ g_{30} & g_{31} & g_{32} & g_{33} \end{pmatrix}^{(x)}$$

g is a symmetric, non-degenerate bilinear norm with signature $(-,+,+,+)$. That is, g has one negative eigenvalue and 3 positive eigenvalues.

$g_{\mu\nu}$ is called metric matrix, its inverse being

$$(g^{\mu\nu}) = (g_{\mu\nu})^{-1}$$

Example 1.4.1 (Polar coordinates)**.**

$$\begin{cases} \eta^0 = x^0 \\ \eta^1 = r \\ \eta^2 = \theta \\ \eta^3 = \varphi \end{cases} = (d\eta^0, d\eta^1, d\eta^2, d\eta^3) \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & r^2 & \\ & & & r^2 \sin^2\theta \end{pmatrix} \begin{pmatrix} d\eta^0 \\ d\eta^1 \\ d\eta^2 \\ d\eta^3 \end{pmatrix}$$

$$\textit{i.e. } (dx^2 + dy^2 + dz^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, dy^2)$$

Assume that M is a 4-D smooth manifold and g is a Lorentz metric on M. Let (U, x^μ) be a local coordinate chart on M (U is an open subset of $\mathbb{R}^4$). Locally g can be written as

$$g = g_{\mu\nu}(x) dx^\mu \otimes dx^\nu$$

Let X be a vector field on U. $X = X^\mu \frac{\partial}{\partial x^\mu}$, where X^μ ($\mu = 0, 1, 2, 3$) are smooth functions on U.

Let $\mathfrak{X}(U) = \{X \mid \text{smooth vector fields on } U\}$.

$$\begin{aligned} g : \mathfrak{X}(U) \times \mathfrak{X}(U) &\to C^\infty(U) \\ (X, Y) \quad &\mapsto g(X, Y) \end{aligned}$$

$$X = X^\mu \frac{\partial}{\partial x^\mu}, \quad Y = Y^\nu \frac{\partial}{\partial x^\nu}$$

Since g is a bilinear function, we can compute it as follows:

$$\begin{aligned} g(X, Y) &= g\left(X^\mu \frac{\partial}{\partial x^\mu}, Y^\nu \frac{\partial}{\partial x^\nu}\right) \\ &= X^\mu Y^\nu g\left(\frac{\partial}{\partial x^\mu}, \frac{\partial}{\partial x^\nu}\right) \\ &= X^\mu Y^\nu g_{\mu\nu} \end{aligned}$$

Q. How to take derivatives on U?

Key properties:

(i) When $g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$, it reduces to the flat case.

(ii) Leibniz rule applies.

Denote by ∇ (仿射联络):

$$\begin{aligned} \nabla : \mathfrak{X}(U) \times \mathfrak{X}(U) &\to \mathfrak{X}(U) \\ (X, Y) &\mapsto \nabla_X Y \end{aligned}$$

注 1.4.1. 其满足如下性质：

(i) $\nabla_{fX+hY} Z = f\nabla_X Z + h\nabla_Y Z$;

(ii) $\nabla_X(fY + hZ) = (Xf)Y + f\nabla_X Y + (Xh)Z + h\nabla_X Z$,

$\nabla_X Y$ 称为 Y 关于 X 的共变导数，或协变导数。

If we take $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}$,

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} := \Gamma_{ij}^k \frac{\partial}{\partial x^k}$$

Torsion free condition: We want $\nabla_X Y - \nabla_Y X = [X, Y]$.

注 1.4.2. 这里 *torsion free* 指无挠，挠率定义为

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

若 $T = 0$，则联络 ∇ 称为无挠联络或对称联络。

Take $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}$:

$$\Gamma_{ij}^k \frac{\partial}{\partial x^k} - \Gamma_{ji}^k \frac{\partial}{\partial x^k} = \left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right] = 0$$

This leads to $(\Gamma_{ij}^k - \Gamma_{ji}^k)\frac{\partial}{\partial x^k} = 0$

Thus $\Gamma_{ij}^k = \Gamma_{ji}^k$.

注 1.4.3. 因此可以看出联络 ∇ 为对称的充要条件是在 M 的局部坐标系下 $\Gamma_{ij}^k = \Gamma_{ji}^k$.

Compatible with the metric (∇ 与度量 g 相容)：

Let X, Y, Z be three vector fields on U.

$$Z(g(X, Y)) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y)$$

We simply denote $g(\cdot, \cdot)$ by $\langle \cdot, \cdot \rangle$

$$\frac{\partial}{\partial x^k}\left(\left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle\right) = \left\langle \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle + \left\langle \frac{\partial}{\partial x^i}, \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^j} \right\rangle$$

i.e.

$$\frac{\partial}{\partial x^k} g_{ij} = \Gamma_{ki}^l g_{lj} + g_{il}\Gamma_{kj}^l. \quad (3)$$

Interchange i and k:

$$\frac{\partial}{\partial x^i} g_{kj} = \Gamma^l_{ik} g_{lj} + \Gamma^l_{ij} g_{kl}. \quad (1)$$

Interchange j and k:

$$\frac{\partial}{\partial x^j} g_{ik} = \Gamma^l_{ji} g_{lk} + g_{il} \Gamma^l_{jk}. \quad (2)$$

$$(1) + (2) - (3) \Rightarrow$$
$$\text{LHS} = \frac{\partial g_{kj}}{\partial x^i} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k}$$

$$\text{RHS} = 2\Gamma^l_{ij} g_{lk} = \text{LHS}$$

Let (g^{ij}) be the inverse matrix of (g_{ij}). (Since we have assumed that g is nondegenerate.) This implies

$$\Gamma^k_{ij} = \frac{1}{2} g^{km} \left(\frac{\partial g_{im}}{\partial x^j} + \frac{\partial g_{mj}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^m} \right)$$

Remark: Γ^k_{ij} is fully determined by the metric g. Now we can define the so-called covariant derivative (协变导数).

Let X and Y be two vector fields on U.

$$X = X^\mu \frac{\partial}{\partial x^\mu}, \quad Y = Y^\nu \frac{\partial}{\partial x^\nu}$$

$$\nabla_X Y = \nabla_{X^\mu \frac{\partial}{\partial x^\mu}} \left(Y^\nu \frac{\partial}{\partial x^\nu} \right) = X^\mu \nabla_{\frac{\partial}{\partial x^\mu}} \left(Y^\nu \frac{\partial}{\partial x^\nu} \right) = X^\mu \left(\frac{\partial Y^\nu}{\partial x^\mu} \frac{\partial}{\partial x^\nu} + Y^\nu \nabla_{\frac{\partial}{\partial x^\mu}} \frac{\partial}{\partial x^\nu} \right)$$

In particular, one considers:

$$\begin{aligned} \nabla_{\frac{\partial}{\partial x^\mu}} \left(Y^\nu \frac{\partial}{\partial x^\nu} \right) &= \frac{\partial Y^\nu}{\partial x^\mu} \frac{\partial}{\partial x^\nu} + Y^\nu \Gamma^\alpha_{\mu\nu} \frac{\partial}{\partial x^\alpha} \\ &= \left(\frac{\partial Y^\alpha}{\partial x^\mu} + Y^\nu \Gamma^\alpha_{\mu\nu} \right) \frac{\partial}{\partial x^\alpha} \end{aligned}$$

We also denote LHS as:

$$\nabla_{\frac{\partial}{\partial x^\mu}} \left(Y^\nu \frac{\partial}{\partial x^\nu} \right) = \nabla_{\frac{\partial}{\partial x^\mu}} \left(Y^\alpha \frac{\partial}{\partial x^\alpha} \right) := (Y^\alpha_{;\mu}) \frac{\partial}{\partial x^\alpha}$$

So we can see

$$Y^\alpha_{;\mu} = \frac{\partial Y^\alpha}{\partial x^\mu} + \Gamma^\alpha_{\mu\nu} Y^\nu$$

Compare LHS and RHS.

注 **1.4.4.** *Mathematician vs physicist notation:*

- *Mathematician:* $Y = Y^\alpha \frac{\partial}{\partial x^\alpha}$; $\nabla_{\frac{\partial}{\partial x^\mu}} Y = Y^\alpha_{;\mu} \frac{\partial}{\partial x^\alpha}$.
- *Physicist:* Y^α; $Y^\alpha_{;\mu} = Y^\alpha_{,\mu} + \Gamma^\alpha_{\mu\nu} Y^\nu$.

  这个式子方便记忆，可以写成：协变导数 = 普通导数 + 联络修正。

Summary: Let (M, g) be a 4-D Lorentz manifold.

There is a unique connection

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$

such that

(i) $\nabla_X Y - \nabla_Y X = [X, Y]$ (torsion-free)

(ii) $Z(g(X, Y)) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y)$ (compatible with metric)

This connection is called the Levi-Civita connection.

注 1.4.5. 事实上，由上面定义，(M, g) 上的一个 *Levi-Civita* 联络是指一个与 g 相容的对称联络。黎曼几何基本定理指出，任一黎曼流形 (M, g) 上存在唯一的 *Levi-Civita* 联络。

Now we work on (U, x^μ), $X = X^\mu \frac{\partial}{\partial x^\mu}$. We denote by $\nabla_\mu X$ the covariant derivative of X along the direction $\frac{\partial}{\partial x^\mu}$, i.e. $\nabla_{\frac{\partial}{\partial x^\mu}} X$

$$(\nabla_\mu X)^\alpha = X^\alpha_{,\mu} + \Gamma^\alpha_{\mu\nu} X^\nu$$

注 1.4.6. 这里使用了物理学家常用的记号：

- $X^\alpha_{,\mu}$ 表示普通偏导数 $\frac{\partial X^\alpha}{\partial x^\mu}$
- $\Gamma^\alpha_{\mu\nu}$ 是克里斯托费尔符号
- $(\nabla_\mu X)^\alpha$ 表示向量场 X 的协变导数的 α 分量

We are going to introduce the curvature operator R.

$$\begin{aligned} R : \mathfrak{X}(U) \times \mathfrak{X}(U) \times \mathfrak{X}(U) &\to \mathfrak{X}(U) \\ (X, Y, Z) &\mapsto R(X, Y)Z \\ &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \end{aligned}$$

In particular, one considers

$$X = \frac{\partial}{\partial x^\alpha} = \partial_\alpha, \quad Y = \frac{\partial}{\partial x^\beta} = \partial_\beta, \quad Z = \frac{\partial}{\partial x^\gamma} = \partial_\gamma$$

Now compute $R(\partial_\alpha, \partial_\beta)\partial_\gamma$.

Define

$$dx^\mu [R(\partial_\alpha, \partial_\beta)\partial_\gamma] \triangleq R^\mu_{\alpha\beta\gamma}$$

注 1.4.7. $R^\mu_{\alpha\beta\gamma}$ 也有别的记号，上下标的写法与这里略有不同。这里为了方便直接这样写，暂时没有产生歧义。

$$\begin{aligned}
R(\partial_\alpha,\partial_\beta)\partial_\gamma &= \nabla_\alpha(\nabla_\beta\partial_\gamma)-\nabla_\beta(\nabla_\alpha\partial_\gamma)-\nabla_{[\partial_\alpha,\partial_\beta]}\partial_\gamma \\
&= \nabla_\alpha(\nabla_\beta\partial_\gamma)-\nabla_\beta(\nabla_\alpha\partial_\gamma) \quad (\text{since } [\partial_\alpha,\partial_\beta]=0) \\
&= \nabla_\alpha(\Gamma^\lambda_{\beta\gamma}\partial_\lambda)-\nabla_\beta(\Gamma^\lambda_{\alpha\gamma}\partial_\lambda) \\
&= \left(\frac{\partial\Gamma^\lambda_{\beta\gamma}}{\partial x^\alpha}\partial_\lambda+\Gamma^\lambda_{\beta\gamma}\Gamma^\rho_{\alpha\lambda}\partial_\rho\right)-\left(\frac{\partial\Gamma^\lambda_{\alpha\gamma}}{\partial x^\beta}\partial_\lambda+\Gamma^\lambda_{\alpha\gamma}\Gamma^\rho_{\beta\lambda}\partial_\rho\right) \text{(Using Leibniz rule)} \\
&= \left(\frac{\partial\Gamma^\rho_{\beta\gamma}}{\partial x^\alpha}-\frac{\partial\Gamma^\rho_{\alpha\gamma}}{\partial x^\beta}+\Gamma^\lambda_{\beta\gamma}\Gamma^\rho_{\alpha\lambda}-\Gamma^\lambda_{\alpha\gamma}\Gamma^\rho_{\beta\lambda}\right)\partial_\rho
\end{aligned}$$

Acting on dx^μ, we get

$$R^\mu_{\alpha\beta\gamma}=\frac{\partial\Gamma^\mu_{\beta\gamma}}{\partial x^\alpha}+\Gamma^\lambda_{\beta\gamma}\Gamma^\mu_{\alpha\lambda}-\left(\frac{\partial\Gamma^\mu_{\alpha\gamma}}{\partial x^\beta}+\Gamma^\lambda_{\alpha\gamma}\Gamma^\mu_{\beta\lambda}\right)$$

$R^\mu_{\alpha\beta\gamma}$ is called the Riemann curvature tensor.

$$R=\partial\Gamma-\partial\Gamma+\Gamma*\Gamma-\Gamma*\Gamma \quad (\text{形式上长的像这个式子，便于记忆})$$

Fact:$R(X,Y)Z$ are C^∞-functions and linear for any C^∞ function f:

$$\begin{aligned}
R(fX,Y)Z &= fR(X,Y)Z \\
R(X,fY)Z &= fR(X,Y)Z \\
R(X,Y)(fZ) &= fR(X,Y)Z
\end{aligned}$$

Assume $X,Y\in\mathfrak{X}(M)$, $w\in A^1(M)$ (1-form)

$$(\nabla_X\omega)(Y)\overset{\text{def}}{=}X(\omega(Y))-\omega(\nabla_XY)$$

Here ω is an 1-form，X and Y are vector fields，$\omega(Y)$ is a C^∞-function.
In local coordinate chart (U,x^μ) ,$\omega=\omega_\mu dx^\mu$.

$$\begin{aligned}
(X^\mu\nabla_\mu\omega_\nu)Y^\nu &= X^\mu\partial_\mu(\omega_\nu Y^\nu)-\omega_\nu(X^\mu\nabla_\mu Y^\nu) \\
&= X^\mu(\partial_\mu\omega_\nu)Y^\nu+X^\mu\omega_\nu\partial_\mu Y^\nu-\omega_\nu X^\mu\partial_\mu Y^\nu-\omega_\nu X^\mu\Gamma^\nu_{\mu\alpha}Y^\alpha \\
&= (\partial_\mu\omega_\nu-\Gamma^\alpha_{\mu\nu}\omega_\alpha)X^\mu Y^\nu
\end{aligned}$$

因此，$\nabla_\mu\omega_\nu=\partial_\mu\omega_\nu-\Gamma^\alpha_{\mu\nu}\omega_\alpha$

注 1.4.8. 这个结果可以推广到高阶张量。例如，对于 $(2,1)$ 型张量，其协变导数为：

$$\nabla_\alpha T^\beta_{\mu\nu}=\partial_\alpha T^\beta_{\mu\nu}+\Gamma^\beta_{\alpha\lambda}T^\lambda_{\mu\nu}-\Gamma^\lambda_{\alpha\mu}T^\beta_{\lambda\nu}-\Gamma^\lambda_{\alpha\nu}T^\beta_{\mu\lambda}$$

An important property: $\nabla g\equiv 0$

$\Rightarrow$ The operations of raising and lowering indices commute with covariant differentiation

$$R^\mu_{\alpha\beta\nu}=dx^\mu(R(\partial_\alpha,\partial_\beta)\partial_\nu)$$

$$R_{\alpha\beta\mu\nu} = g_{\mu\rho}R^{\rho}_{\alpha\beta\nu}$$

$$\begin{aligned}(g_{ij})_{;k} &= \frac{\partial g_{ij}}{\partial x^k} - \Gamma^{\lambda}_{kj}g_{i\lambda} - \Gamma^{\lambda}_{ki}g_{\lambda j}\\ &= 0 \quad \text{(compatible with the metric)}\end{aligned}$$

注 1.4.9. 由前面对张量做协变导数的说明,对任何向量场 Z, 协变导数 $\nabla_Z g$ 是有意义的。若 $\nabla g = 0$, 即对任意向量场 Z, 都有 $\nabla_Z g = 0$, 则称联络 ∇ 与黎曼度量 g 是相容的。$\nabla g = 0$ 等价于 $(g_{ij})_{;k} = 0$, 具体计算在上面，对 g 的协变导数计算可以仿照 1-*form* 和 $(2,1)$ 张量的方式给出。

Now we do some calculation in a coordinate basis.

$$X = X^{\alpha}\partial_{\alpha} \quad Y = Y^{\beta}\partial_{\beta} \quad Z = Z^{\nu}\partial_{\nu}$$

$$\begin{aligned}R(X,Y)Z \quad (\text{ a vector field}) &= R(X^{\alpha}\partial_{\alpha}, Y^{\beta}\partial_{\beta})(Z^{\nu}\partial_{\nu})\\ &= X^{\alpha}Y^{\beta}Z^{\nu}R(\partial_{\alpha},\partial_{\beta})\partial_{\nu} \quad (\text{Since} \quad C^{\infty}\text{-linear})\\ &= [R^{\mu}_{\alpha\beta\nu}X^{\alpha}Y^{\beta}Z^{\nu}]\frac{\partial}{\partial x^{\mu}}\end{aligned}$$

$R^{\mu}_{\alpha\beta\nu}X^{\alpha}Y^{\beta}Z^{\nu}$(表示$R(X,Y)Z$ 的第μ 个分量)

$$\begin{aligned}&= X^{\alpha}\nabla_{\alpha}(Y^{\beta}\nabla_{\beta}Z^{\mu}) - Y^{\alpha}\nabla_{\alpha}(X^{\beta}\nabla_{\beta}Z^{\mu}) - (X^{\alpha}\partial_{\alpha}Y^{\beta} - Y^{\alpha}\partial_{\alpha}X^{\beta})\nabla_{\beta}Z^{\mu}\\ &= (\text{Leibniz})X^{\alpha}(\nabla_{\alpha}Y^{\beta})\nabla_{\beta}Z^{\mu} + X^{\alpha}Y^{\beta}\nabla_{\alpha}\nabla_{\beta}Z^{\mu}\\ &- Y^{\alpha}(\nabla_{\alpha}X^{\beta})\nabla_{\beta}Z^{\mu} - Y^{\alpha}X^{\beta}\nabla_{\alpha}\nabla_{\beta}Z^{\mu}\\ &- (X^{\alpha}\partial_{\alpha}Y^{\beta})\nabla_{\beta}Z^{\mu} + (Y^{\alpha}\partial_{\alpha}X^{\beta})\nabla_{\beta}Z^{\mu}\\ &= X^{\alpha}Y^{\beta}(\nabla_{\alpha}\nabla_{\beta} - \nabla_{\beta}\nabla_{\alpha})Z^{\mu}.\end{aligned}$$

注 1.4.10. 刚刚是使用 R 的定义，即 $R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$. 从而按分量展开计算:

$$\nabla_X\nabla_Y Z = X^{\alpha}\nabla_{\alpha}(\nabla_Y Z) = X^{\alpha}\nabla_{\alpha}(Y^{\beta}\nabla_{\beta}Z),$$

$$[X,Y]f = X(Yf) - Y(Xf) = (X^{\alpha}\partial_{\alpha}Y^{\beta} - Y^{\alpha}\partial_{\alpha}X^{\beta})\partial_{\beta},$$

$$\nabla_{[X,Y]}Z = \nabla_{(X^{\alpha}\partial_{\alpha}Y^{\beta} - Y^{\alpha}\partial_{\alpha}X^{\beta})\partial_{\beta}}Z = (X^{\alpha}\partial_{\alpha}Y^{\beta} - Y^{\alpha}\partial_{\alpha}X^{\beta})\nabla_{\beta}Z$$

On the other hand,

$$R^{\mu}_{\alpha\beta\nu}Z^{\nu} = (\nabla_{\alpha}\nabla_{\beta} - \nabla_{\beta}\nabla_{\alpha})z^{\mu}$$

$$2\nabla_{[\alpha}\nabla_{\beta]}z_{\mu} = R_{\alpha\beta\mu\nu}z^{\nu}$$

where the square brackets indicate anti-symmetricization

$$T_{[\alpha\beta]} = \frac{1}{2}(T_{\alpha\beta} - T_{\beta\alpha})$$

This is readily generalized for arbitrary tensors

$$2\nabla_{[\alpha}\nabla_{\beta]}(z_\mu\omega_\nu) = 2(\nabla_{[\alpha}\nabla_{\beta]}z_\mu)\omega_\nu + 2(\nabla_{[\alpha}\nabla_{\beta]}\omega_\nu)z_\mu$$
$$= R_{\alpha\beta\mu\sigma}z^\sigma\omega_\nu + R_{\alpha\beta\nu\sigma}\omega^\sigma z_\mu$$

One concludes

$$2\nabla_{[\alpha}\nabla_{\beta]}T_{\mu\nu} = R_{\alpha\beta\mu\sigma}T^\sigma_\nu + R_{\alpha\beta\nu\sigma}T^\sigma_\mu$$

Let $Z_\mu = \nabla_\mu f = \partial_\mu f$ where f is a smooth function.

$$R_{[\alpha\beta\mu]\nu}Z^\nu = 2\nabla_{[[\alpha}\nabla_{\beta]}Z_{\mu]} = 2\nabla_{[\alpha}\nabla_{[\beta,}Z_{\mu]]} = 0$$

because $\nabla_{[\mu}, Z_{\nu]} = \partial_{[\mu}, Z_{\nu]} - \Gamma^\alpha_{[\mu\nu]}Z_\alpha = \partial_{[\mu}, \partial_{\nu]}f = 0$
Since z can be chosen freely,

$$R_{[\alpha\beta\mu]\nu} = 0 \iff R_{\alpha\beta\mu\nu} + R_{\beta\mu\alpha\nu} + R_{\mu\alpha\beta\nu} = 0$$

That is

$$R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0$$

It is called the first Bianchi identity.

$$T_{[\alpha\beta\gamma]} = \frac{1}{6}[T_{\alpha\beta\gamma} + T_{\beta\gamma\alpha} + T_{\gamma\alpha\beta} - T_{\beta\alpha\gamma} - T_{\alpha\gamma\beta} - T_{\gamma\beta\alpha}]$$

注 1.4.11. 事实上，可以如下直接证明 *Bianchi* 第一恒等式.
由挠率为 0，我们有 $\nabla_X Y - \nabla_Y X = [X,Y]$. 从而有：

$$\begin{aligned}
&R(X,Y)Z + R(Y,Z)X + R(Z,X)Y \\
&= \nabla_X\nabla_Y Z - \nabla_Y\nabla_X - \nabla_{[X,Y]}Z + \cdots \\
&= \nabla_X(\nabla_Y Z - \nabla_Z Y) - \nabla_{[Y,Z]}X + \cdots \\
&= \nabla_X[Y,Z] - \nabla_{[Y,Z]}X + \cdots \\
&= [X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0
\end{aligned}$$

其中 $\cdots$ 后面都是和前面轮换对称的形式。最后一步是由于 *Jacobi* 恒等式。

The 2nd Bianchi identity：

$$R_{\alpha\beta\mu\nu} = R^\lambda_{\alpha\beta\nu}g_{\mu\lambda}$$

$$R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu} = -R_{\alpha\beta\nu\mu} = R_{\mu\nu\alpha\beta}$$

$$\nabla_\alpha R_{\beta\gamma\mu\nu} + \nabla_\beta R_{\gamma\alpha\mu\nu} + \nabla_\gamma R_{\alpha\beta\mu\nu} = 0$$

$$\Leftrightarrow \nabla_{[\alpha}R_{\beta\gamma]\mu\nu} = 0$$

Define Ricci tensor:

$$R_{\mu\nu} = \sum_{\alpha} R^{\alpha}_{\alpha\mu\nu},$$

and Scalar curvature:

$$R = g^{\mu\nu} R_{\mu\nu},$$

where $g^{\mu\nu}$ is the inverse of $g_{\mu\nu}$.

By contracting the pairs of indices (β, μ) and (γ, ν) in the second Bianchi identity,

$$\nabla_\alpha R - \nabla^\beta R_{\alpha\beta} - \nabla^\gamma R_{\alpha\gamma} = 0$$

That is

$$\nabla^\beta R_{\alpha\beta} - \frac{1}{2}\nabla_\alpha R = 0 \iff \nabla^\beta (R_{\alpha\beta} - \frac{1}{2} R g_{\alpha\beta}) = 0$$

注 1.4.12. 这里的记号含义如下：

$$\nabla_\mu = \frac{\partial}{\partial x^\mu},$$
$$\nabla^\mu = g^{\mu\nu}\nabla_\nu$$

Now we introduce the so-called Einstein-tensor

$$G_{\mu\nu} := R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \quad \text{which is divergence-free, that is,}$$
$$\nabla^\nu G_{\mu\nu} = 0.$$

The equation $G_{\mu\nu} = 0$ is called the vacuum Einstein equation.

Remark: (VEE) is equivalent to $R_{\mu\nu} = 0$.

证明. $\Leftarrow$: is trivial.

$\Rightarrow$:

$$\begin{aligned} & R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0 \\ \Rightarrow\ & g^{\mu\nu} R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} g^{\mu\nu} = 0 \\ \Rightarrow\quad & R - \frac{1}{2} R \cdot 4 = 0 \quad \Rightarrow \quad R = 0 \quad \Rightarrow R_{\mu\nu} = 0 \end{aligned}$$

□

The Minkowski metric $\eta = -(dx_0)^2 + \sum_{i=1}^{3} (dx_i)^2$ obviously satisfies the VEE. (平直时空曲率张量为 0)